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The effect of relative crest submergence on wave breaking over submerged slopes

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ABSTRACT

Experiments were performed in a wave flume to measure the intensity, transmission and reflection of waves breaking over a submerged reef with an offshore gradient of 1:10. The results demonstrate that the relative water depth over the reef crest (h_c/H_o) is a dominant factor affecting the breaking characteristics. In particular it is found that as the relative crest submergence is reduced, there is a considerable increase in the intensity of wave breaking over the reef that can be quantified through measurements of the air cavity enclosed beneath the plunging jet. It is also shown that there is a corresponding decrease in wave transmission and reflection as the submergence is reduced.

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1. Introduction

The wave breaking process drives many coastal processes and so knowledge of the properties of breaking waves is essential if accurate predictions of beach evolution, wave forces on coastal structures, nearshore currents and set up at the shoreline are to be obtained. With this in mind, many theoretical, experimental and numerical studies have been completed on various aspects of the wave breaking process, but the mechanics of wave breaking are not fully understood and there remains no complete theoretical description of wave breaking.

Because existing wave theories become invalid as waves approach breaking, the majority of current knowledge about the characteristics of breaking waves has been obtained from experimental studies. In particular, many authors including Iversen (1952), Goda (1970), Singamsetti and Wind (1980), Sunamura (1981), Galvin (1969) and Weggel (1972) have made measurements of waves breaking on emergent, planar slopes in two-dimensional laboratory wave flumes to determine various properties such as the breaker type, breaker depth index (ratio of breaking wave height to water depth) and breaker height index (ratio of breaking to deepwater wave height). The empirical relationships obtained from these studies are commonly used in shoreline evolution models to define the wave parameters at breaking.

It was noted by Smith and Kraus (1991) that while plane slopes provide a reasonable first approximation of beach shape, natural beach profiles are often more complex and breaking commonly takes place over offshore reefs or breakpoint bars rather than directly on the beach face. In addition, the use of low profile structures such as submerged breakwaters or artificial reefs that provide shoreline protection by causing waves to break offshore are becoming more popular. Consequently, to optimise the design of submerged coastal structures and for accurate prediction of coastal processes it is important to determine any differences between wave breaking on emergent and submerged slopes.

Smith and Kraus (1990, 1991, 1992) recognised the need to examine wave breaking over submerged structures and carried out a comprehensive series of laboratory measurements on submerged bar/reef structures with various geometries. The results demonstrated that waves with identical deepwater characteristics break differently on plane beaches and bars/reefs. Significant differences were found in properties such as breaker type, breaker height index, breaker depth index, plunge distance (the distance from initial breaking to the "plunge point", where the overturning wave jet impacts the water surface in the preceding wave trough) and splash distance (the distance from the plunge point to the "splash point" where the primary splash generated by impact of the overturning jet returns to the water surface). In particular, it was found that the transition values of the surf similarity parameter for the classification of breaker type suggested by Battjes (1974) were invalid for barred profiles, and waves of moderate steepness that plunge on plane slopes tended to collapse when breaking on a bar/reef structure.

One of the major differences between a submerged bar/reef and a plane slope is that a bar/reef has a fixed crest level and thus in tidal locations, the relative submergence of the crest h_c/H_o where h_c is the water depth above the crest and H_o is the deepwater wave height

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varies over the course of the tidal cycle and this may have an effect on the nature of wave breaking. Anecdotal evidence to support this suggestion comes from the observations of surfers that waves breaking over a submerged reef become "hollower", i.e. breaking in a more plunging manner, as the tide level falls and the water depth over the reef crest is reduced. This variation in wave breaking intensity with water level was noted by Sayce et al. (1999) who suggested that for waves breaking on submerged reefs, the breaker type becomes more strongly dependent on wave height and water depth, rather than wave steepness and beach gradient as suggested by Iversen (1952), Galvin (1968) and Battjes (1974) for plane slopes. Gourlay (1994) examined the transformation of regular waves breaking on a laboratory model of a fringing reef with a steep seaward face and a gradually sloping reef top. His results demonstrated that breaking conditions were affected by the relevant submergence on the reef and waves with identical deepwater steepness ($H_o/L_o = 0.0408$) broke in a plunging manner when $h_c/H_o \leq 1.8$ and by spilling when $h_c/H_o \geq 1.8$.

Further evidence that demonstrates the importance of water depth over the crest of a submerged structure to the nature of wave breaking can be found in existing literature on submerged breakwaters. Wave transmission past both submerged and low crested breakwaters has been examined by many researchers including Ahrens (1987), Seelig (1980), van der Meer (1990), Allsop (1983) and Friebel and Harris (2004), and was found to be dependent on the relative water depth over the reef crest, incident wave conditions, relative crest width and the nominal diameter of the armourstone used for breakwater construction. In particular the transmitted wave height showed strong dependence on the relative submergence of the crest and decreased as the water depth over the reef crest was reduced. As the majority of wave height decay occurs due to energy dissipation during wave breaking, these measurements suggest that the nature of the breaking event must vary as the relative submergence of the reef is changed.

It is noted that while Smith and Kraus (1990, 1991, 1992) completed by far the most comprehensive measurements of wave breaking over submerged structures to date, they maintained a water depth over the reef crest h_c of approximately $0.66H_b$ during their experiments and consequently the effect of varying water level on the breaker characteristics was not examined. The present paper describes a series of laboratory experiments designed to investigate the influence of the water level on the properties of waves breaking over submerged structures.

2. Experimental arrangements

The experiments described in this paper were performed in a glass-walled wave flume at the University of Southampton. The wave tank measures 17 m in length, 0.44 m wide and was filled with tap water to a depth of 0.7 m (Fig. 1). The flume is equipped with a flap-type wave generator which allows active absorption of reflections. The far end of the flume is fitted with a vertical wedge of polyether foam designed to absorb incoming waves; the reflection coefficient was shown to be between 2% and 8% for the wave conditions measured in the present experiments.

A submerged reef was constructed using high-density 15 mm thick plastic and installed in the flume with the toe of the reef 3.8 m from the wavemaker. The reef has a height of 618 mm and a front gradient of 1:10. On the rear side it drops away vertically from the crest. This geometry was used so that regular waves would break in depth limited conditions close to the crest of the reef and reform almost immediately in the deepwater behind in an attempt to isolate the initial wave breaking process. Water surface elevations were recorded with resistance-type wave gauges at a rate of 256 samples per wave period, at 7 locations shown in Fig. 1. The first two gauges were installed in front of the submerged reef structure in order to measure the incident waves. A further five wave gauges were installed beyond the reef to measure the transmitted waves which reform in the deep water section of the flume after breaking. There were two wave gauges at 11.5 m from the wavemaker, with a transverse separation of 0.2 m, to detect any cross-tank sloshing in the flume during the tests. This was found to be negligible.

A PULNIX TM-6710 monochrome digital video camera as was installed alongside the wave flume to record images of waves breaking over the crest of the reef. The camera was aligned perpendicular to the wave flume and was moved horizontally and vertically as the water level and breaker position varied in order to best capture the wave breaking process. Four 500 W spotlights were installed on the far side of the flume opposite the camera position, behind a translucent screen. This lighting set-up allowed shutter speeds to be as high as 1/32,000 s, providing well defined images of the wave breaking process. A grid with 10 mm vertical intervals and 30 mm horizontal intervals was attached to the glass on the camera's side of the wave flume as a reference scale.

3. Results

3.1. Breaker indices

This section describes a series of measurements designed to obtain breaker height and breaker depth indices for waves breaking over the submerged reef, and their dependence on the water depth over the crest of the reef.

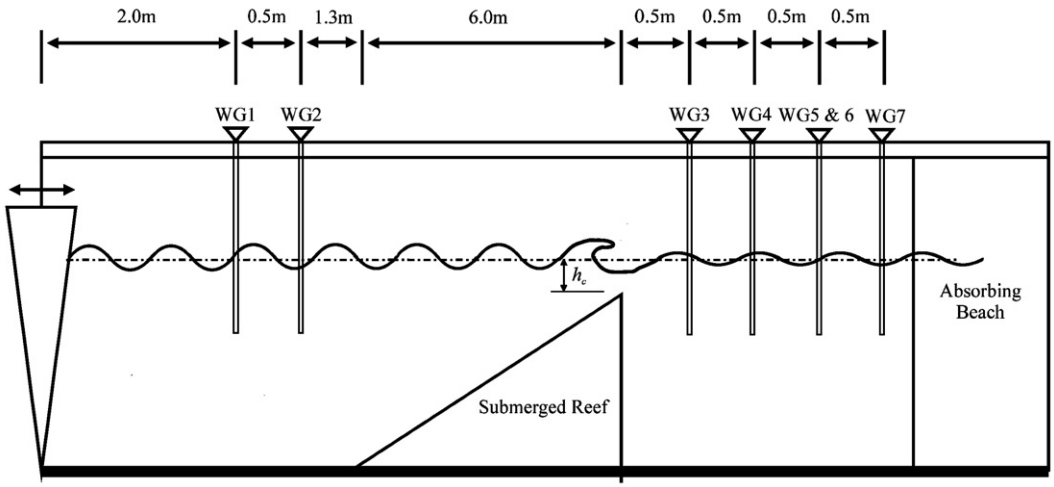


Fig. 1. Experimental arrangement.

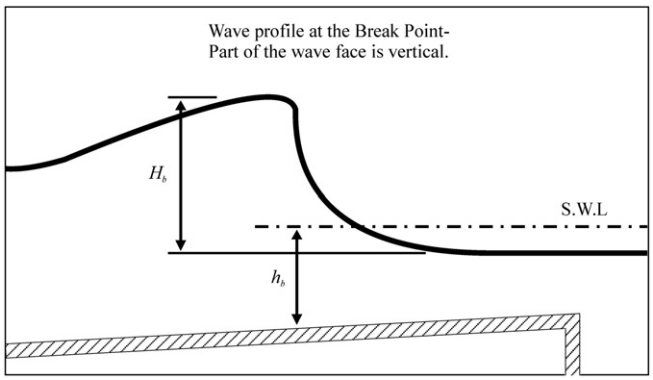


Fig. 2. Definition wave profile parameters at the break point.

3.1.1. Breaker depth index

The breaker depth index γ_b is used to determine the break point in nearshore wave models and is important for the calculation of forces on coastal structures in shallow areas. The index is defined as

$$\gamma_b = \frac{H_b}{h_b}, \tag{1}$$

where H_b and h_b are the wave height and water depth at the break point respectively, as shown in Fig. 2.

A large number of studies (Munk, 1949; Collins and Weir, 1969; Singamsetti and Wind, 1980; Sunamura, 1981; Komar and Gaughan, 1973; Galvin, 1969; Weggel, 1972) have reported on the breaker depth index on plane slopes, but only Smith and Kraus (1990, 1991, 1992) seem to have examined γ_b for waves breaking over submerged reefs.

For the breaker index tests, the total water depth in the wave flume was kept constant at 0.7 m, and 48 different wave cases with frequencies ranging from 0.4 Hz to 1.0 Hz and offshore wave heights H_o in the range 40 mm to 105 mm were examined, corresponding to relative submergences h_c/H_o in the range 0.76 to 2. The position of the break point and the magnitude of H_b and h_b were determined through analysis of digital video images of wave breaking for each case. The break point was defined according to the description of Iversen (1952) as the point where any portion of the forward face of the wave first becomes vertical (Fig. 2). This definition is consistent with the majority of breaking wave studies.

Wave breaking is a highly unstable process which can be sensitive to small variations in factors such as the water level or the presence of reflected waves. In addition, the measurement of the wave geometry at breaking and determination of the break point involves a degree of subjectivity and this tends to introduce scatter into the results. To reduce the level of scatter, at least 5 waves were measured for each test case to define an average value for each quantity.

Breaker depth indices are plotted in Fig. 3(a) as a function of relative crest submergence. While there is some scatter in the present results, γ_b maintains an approximately constant value of 0.85 and there is no apparent dependence of γ_b on the relative submergence of the reef crest. Fig. 3(b) plots the results as a function of deepwater wave steepness H_o/L_o and also shows data from several previous experiments (McCowan, 1891; Smith and Kraus, 1991; Singamsetti and Wind, 1980; Sunamura, 1981; Munk, 1949; Komar and Gaughan, 1973; Galvin, 1969; Collins and Weir, 1969; Weggel, 1972), all of which except Smith and Kraus (1991) were made on emergent beach slopes. Unlike most previous measurements, the breaker depth indices measured in the current tests appear to be insensitive to deepwater wave steepness and the closest agreement is with the commonly used breaker criterion of McCowan (1891), $\gamma_b = 0.78$. Surprisingly, there are considerable differences between the measured breaker depth indices and those of Smith and Kraus (1991).

3.1.2. Breaker height index

The breaker height index is defined as

$$\Omega_b = \frac{H_b}{H_0}, \tag{2}$$

and has been shown by previous studies on emergent, planar slopes to be dependent on both offshore wave steepness H_o/L_o and seabed gradient m (Munk, 1949; Singamsetti and Wind, 1980; Sunamura, 1981; Komar and Gaughan, 1973; Weggel, 1972).

Breaker height indices Ω_b are shown as a function of relative crest submergence in Fig. 4(a). It is observed that the data shows some scatter but a weak dependence of Ω_b on relative crest submergence is apparent. A stronger relationship is found if the measured breaker height indices are displayed as a function of deepwater wave steepness (Fig. 4(b)). The present data (for $0.76 < h_c/H_o < 2$) and those of Smith and Kraus (1991, for $h_c/H_o = 0.66$) show a strong trend of increasing breaker height index with decreasing wave steepness. This trend is typical of waves breaking on a plane slope due to the greater shoaling experienced by longer period waves before breaking. The

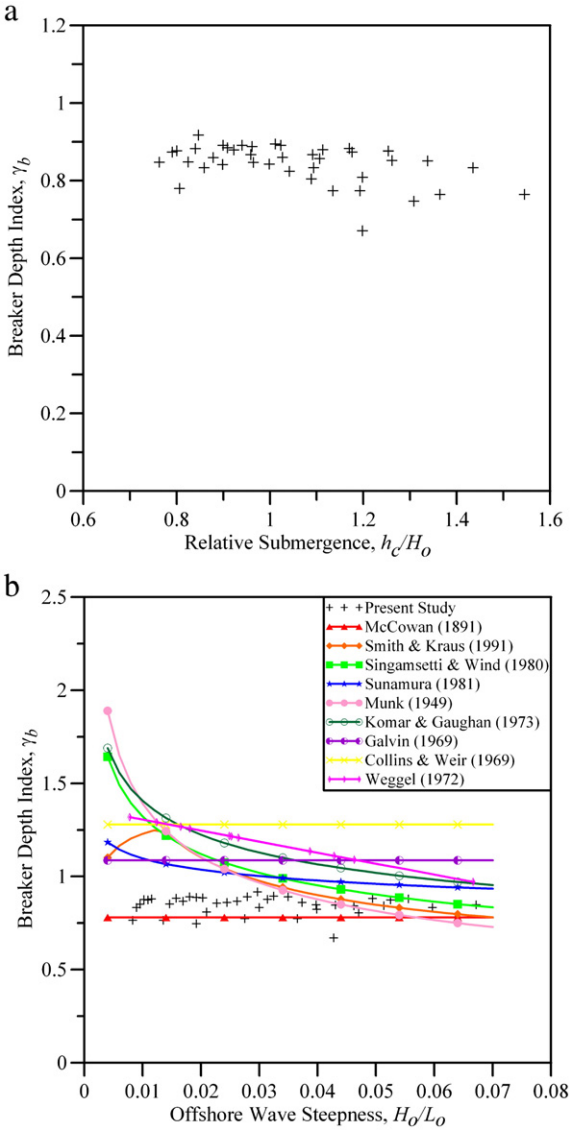


Fig. 3. (a) Measured breaker depth indices as a function of relative crest submergence, and (b) deepwater wave steepness. All except for the present data and those from Smith and Kraus (1991) are for emergent plane slopes.

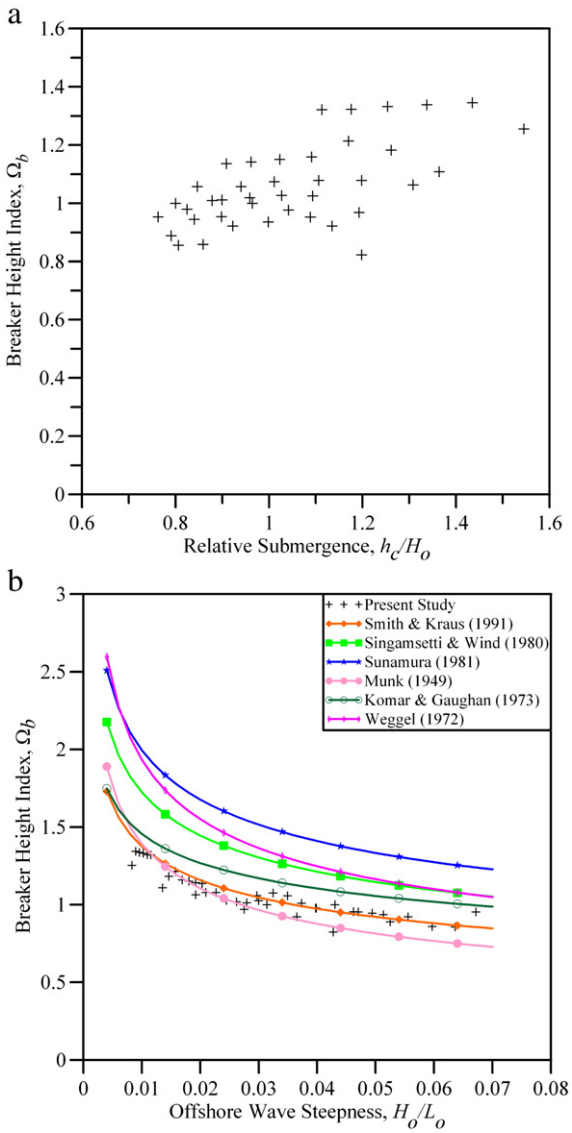


Fig. 4. (a) Measured breaker height index as a function of relative crest submergence, and (b) deepwater wave steepness. All except for the present data and Smith and Kraus (1991) are for emergent plane slopes. Eq. (3) is plotted as a dashed line.

work of previous researchers has shown that the data typically follows a power law and can be expressed in the form

$$\Omega_b = C(m)\left(\frac{H_0}{L_0}\right)^{-n(m)}. \tag{3}$$

However as only one reef gradient is examined in the present study, the values of $C(m)$ and $n(m)$ are constant and for the current case have values of 0.5 and 0.2 respectively.

The breaker height indices obtained by previous authors are also shown on Fig. 4(b). It is seen that the majority of the present results fall at the low end of the range of these values, and it is encouraging that the data compares most favourably with the data of Smith and Kraus (1991) as this is the only other dataset that was obtained from waves breaking on submerged reefs rather than emergent planar slopes. Smith and Kraus (1991) noted that the presence of an offshore current produced by the high velocity backwash from a preceding wave on an emergent slope would act to increase wave heights at breaking. If it is assumed that the shoreward mass flux in the overturning wave crest must be balanced by the mass flux in the seaward return flow, the velocity of the return flow will increase as the relative crest submergence is reduced and the situation approaches the emergent beach case. Thus in the current case of a submerged reef, the velocity of the backwash is expected to be lower than that on an emergent slope and the effect of the flow on the breaking event is reduced, leading to lower values of Ω_b as seen in Fig. 4(b). The variation of return flow velocity with crest submergence is examined further in Section 3.2.2.

3.2. Breaking intensity

This section describes a series of experiments completed to address the suggestion made in Section 1, that waves break in a more intense manner as the relative submergence of the reef crest is reduced. The common method for defining breaker type is to describe breaking in terms of the four categories suggested by Galvin (1968), i.e. spilling, plunging, collapsing or surging. For present purposes however, these terms are rather too subjective and each category spans a wide range of different breaking wave shapes. Consequently, in order to examine the effect of relative reef submergence it was necessary to define some measurable parameters that could be used to quantify the observed variation in wave breaking intensity.

Mead and Black (2001) also found that the breaker type categories of Galvin (1968) provided an imprecise description of surfing waves. For a more quantitative indication of breaker type they made use of Longuet-Higgins's (1982) parametric solution of the equations of motion. This shows that a good description of the forward face of an overturning wave is provided by a curve of the form

$$\frac{y}{w_c} = \pm\frac{3\sqrt{3}}{4}\sqrt{\frac{x}{l_c}}\left(\frac{x}{l_c}-1\right). \tag{4}$$

where l_c and w_c are the length and width respectively of the cavity of air enclosed between the overturning jet and the wave face and the x and y axes suitably aligned with the arrows in Fig. 5. From photographs of breaking waves at 23 international surfing breaks, Mead and Black observed that those on steep reefs broke in a more violent plunging manner, and that there was an inverse linear relationship between the length to width ratio of the cavity l_c/w_c and the orthogonal reef gradient (for present purposes the same as the actual seabed gradient m), independent of the incoming wave parameters:

$$\frac{l_c}{w_c} = \frac{0.065}{m} + 0.821. \tag{5}$$

In the present experiments, the influence of relative reef submergence on the cavity shape was examined for 7 wave cases

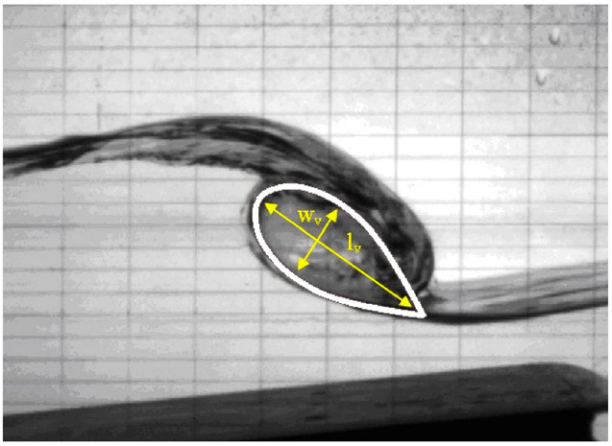


Fig. 5. Photograph of a laboratory wave at the plunge point with a cubic curve fitted to the cavity shape (f=0.6 Hz, H_o=80 mm).

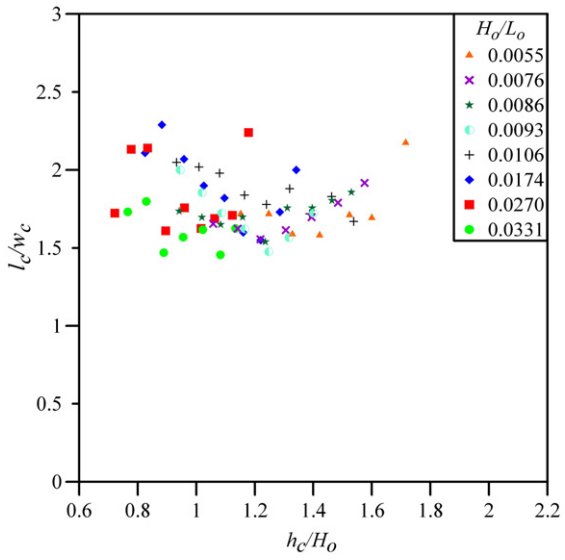


Fig. 6. Measured cavity length to width ratio as a function of relative water depth over the reef crest.

with deepwater wave steepnesses in the range $0.0055 < H_o/L_o < 0.0331$, run at 9 different water levels (62 mm $< h_c <$ 102 mm). The cavity lengths and widths were measured by digitising points on scaled digital images of waves at the plunge point (Fig. 5). As these measurements require the existence of a clearly defined cavity, only waves that fell within the plunging wave category were examined. In many cases some wall effects which occurred principally due to friction between the high velocity plunging jet and the side walls of the flume were evident and in some cases made determination of the dimensions difficult. However by using high contrast and brightness levels when analysing the images it was possible to determine the predominant cavity shape in the majority of cases, with accuracy on length of approximately ±10%. To ensure that consistent results were obtained, an average of the measurements from 5 waves for each test case was taken.

Fig. 6 shows the measured cavity length to width ratio as a function of the relative crest depth h_c/H_o. Values of l_c/w_c varied between 1.46 and 2.28 but do not suggest any strong dependence on the water depth over the reef crest. All the measured ratios are within the range of those observed by Mead and Black (2001) and Sayce et al. (1999) in the field, and by Couriel et al. (1998) in the laboratory, shown in Table 1. The mean of all the measured cavity ratios is 1.76, which is of the same order as the value of 1.47 calculated from Eq. (5) with $m = 0.1$. It is close to the figure of 1.73 computed by New (1983) for the aspect ratio of an elliptical solution of the shape of the front face of plunging wave, but is considerably lower than the theoretical result of 2.76 given by Longuet-Higgins (1982).

While no systematic relationship between the cavity length to width ratio and water level could be established from the current data, a better correlation was found between the cavity area A_c, normalised by the square of the breaking wave height H_b, and the relative water depth over the reef crest (Fig. 7). The area enclosed by the curve (5) is

$$A_c = \frac{2\sqrt{3}}{5} l_c w_c. \tag{6}$$

The cavity areas plotted in Fig. 7 were computed in this way from measured values of l_c and w_c for each test case. Errors are in the region of ±20%.

Although there is considerable scatter, Fig. 7 shows that for a given deepwater steepness, the area of the cavity increases in relation to the square of the deepwater wave height, as the relative water depth over the reef crest decreases. In other words, a wave with given deepwater conditions becomes more plunging in nature as the water depth over the reef crest is reduced and the situation approaches the emergent slope case. The reason for this observed variation may be explained by previous suggestions that interaction between the incoming breaker and the return flow from the previous wave augments the velocity asymmetry in the incoming breaker and promotes a higher breaking intensity as the relative submergence reduces and the velocity of the return flow increases (Iversen, 1952; Weishar and Byrne, 1978; Smith and Kraus, 1990). This is discussed further in Section 3.2.2.

The data presented in Fig. 7 also shows a trend of increasing relative cavity size with decreasing deepwater steepness. This can be partially explained by the fact that longer period waves shoal to a greater height before breaking.

Battjes (1974) showed that for an emergent plane slope, the breaker type could be predicted using a surf similarity parameter $\xi_o = m/(H_o/L_o)^{0.5}$, and suggested transition values of ξ_o for each of Galvin's (1968) categories. Smith and Kraus (1990, 1991) found that incident waves with identical deepwater characteristics break differently on plane beaches and submerged slopes and suggested modified transition values of ξ_o. The results presented here provide quantitative evidence to support the observation that a wave with identical deepwater characteristics will break in a more plunging manner as the water depth over the reef crest is reduced. This observed dependence of the breaker shape on the water level over the reef crest suggests that Battjes's surf similarity parameter in its current form may be unsuitable for the prediction of breaker type on submerged structures, and would have to be extended to take the relative reef submergence h_c/H_o into account.

Table 1
Table showing measured values of the cavity length to width ratios in breaking waves

Study	Type of experiment	Max l_c/w_c	Min l_c/w_c
Couriel et al. (1998)	Laboratory	3.4	1.5
Sayce et al. (1999)	Field	4.01	1.79
Mead and Black (2001)	Field	3.43	1.42
Present study	Laboratory	2.29	1.46

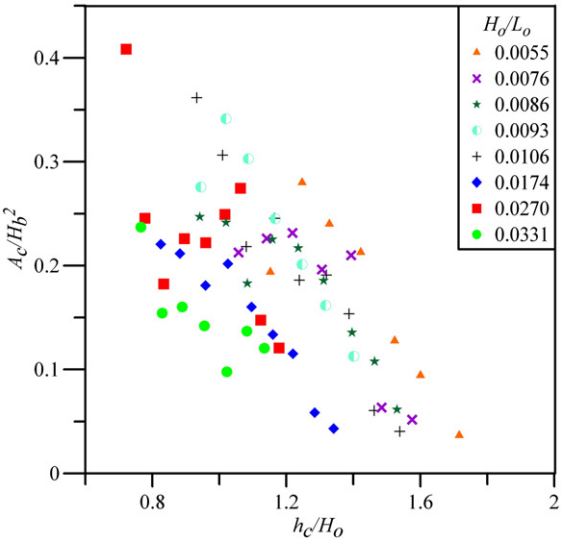


Fig. 7. Cavity areas normalised by the square of the breaking wave height as a function of relative water depth over the reef crest, h_c/H_o.

3.2.1. Energy dissipation

Wave breaking is associated with the generation of high levels of turbulence, air entrainment, noise and splash, all of which must contribute to the energy dissipation and which are seen to increase with wave breaking intensity as waves become more plunging in nature (Blenkinsopp and Chaplin, 2007; Govender et al., 2002; Yasuda et al., 1999). In order to investigate this apparent relationship between breaking intensity and energy dissipation, Fig. 8 shows the energy dissipated by each wave breaking event as a function of the relative cavity area, where the values for the energy dissipated during initial wave breaking were derived from the wave transmission results presented in Section 3.3, taking into account the energy reflected from the reef face.

Although there is some scatter, this figure appears to confirm the existence of a relationship between breaking intensity and initial energy dissipation, showing that the total energy dissipated in the breaking event for each wave case increases as the relative cavity area becomes larger, i.e. as the intensity of breaking increases. These results are in agreement with those of Loewen and Melville (1994) and Lamarre (1993), who found a similar strong correlation between the area of the cavity A_c and the energy dissipated at breaking.

3.2.2. Return flow velocities

It was shown in Section 3.2 that the intensity of wave breaking increases as the relative submergence of the reef crest is reduced, leading to increased air entrainment (Blenkinsopp and Chaplin, 2007) and energy dissipation. This is an interesting result as previous studies on emergent, planar slopes suggested that waves could be well described using only the surf similarity parameter which is dependent solely on the deepwater steepness and beach gradient (Battjes, 1974). As noted above, previous investigators have suggested that interaction between the incoming breaker and the return flow from the previous wave influences the intensity of breaking and this may at least partially explain the observed variations.

In order to investigate this effect, measurements of the horizontal particle velocity over the crest of the reef were made using a TSI fibre optic based LDV system. Horizontal velocity measurements were made for a single wave case with a frequency of 0.8 Hz and an offshore wave height of 80 mm at four different water depths (h_c=67, 72, 82 and 92 mm). Velocity measurements were taken at 5 mm vertical intervals from a level 1 mm above the bed to within 5 mm of the trough level.

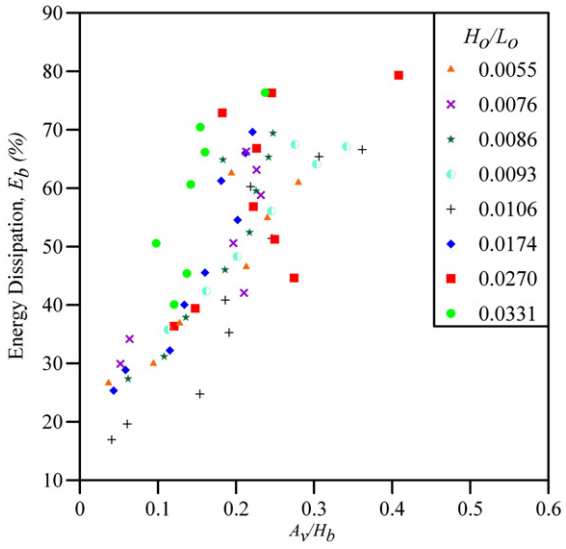


Fig. 8. Wave energy dissipation as a function of relative cavity area.

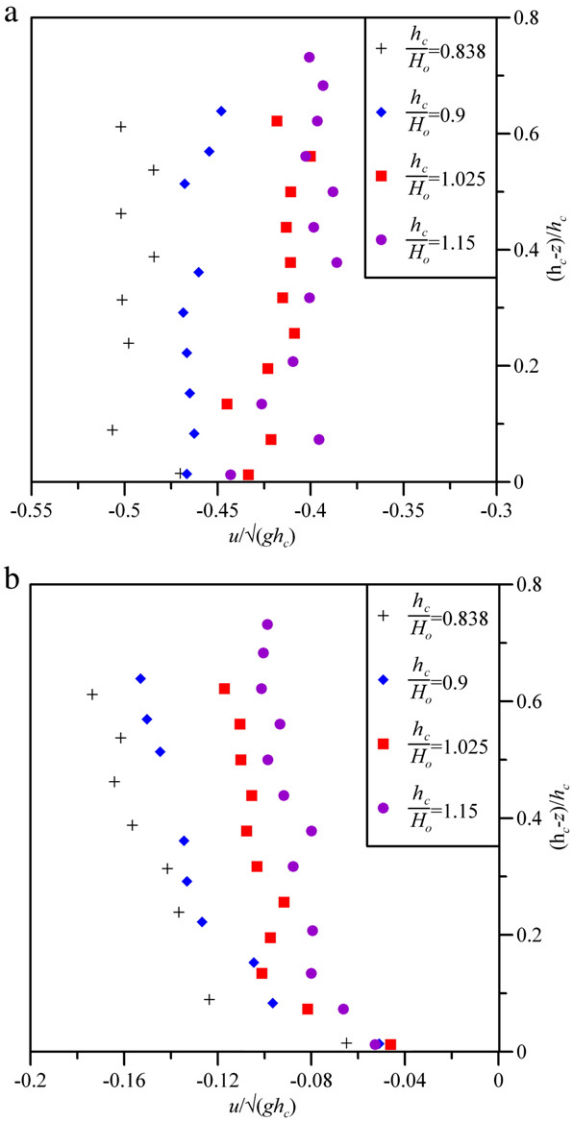


Fig. 9. (a) Maximum return flow velocity, and (b) mean flow velocity averaged over a wave period at each measurement elevation.

The results are plotted in Fig. 9(a) and (b) in the form of maximum return flow velocities and velocities averaged over a wave period at each measurement elevation. The mean flow rate per metre width beneath the trough level over ten waves is shown in Table 2, this was very similar for the four water levels under examination, and was presumably balanced by the flow (not measured) in the overturning wave crest. As the submergence of the reef crest is reduced, the velocity of the return flow increases, and it is reasonable to link this with the observed increase in the breaking intensity. This trend of increasing breaking intensity as expected to continue as the value of h_c/H_o reduces further. However as h_c/H_o approaches zero and then becomes negative, the nature of the flow will fundamentally change.

Table 2
Table showing the mean flow rate below the trough level over ten waves

h_c (mm)	h_c/H_o	Mean flow rate, q (m^3/s/m)
67	0.8375	−0.0061
72	0.9	−0.0064
82	1.025	−0.0059
92	1.15	−0.0061

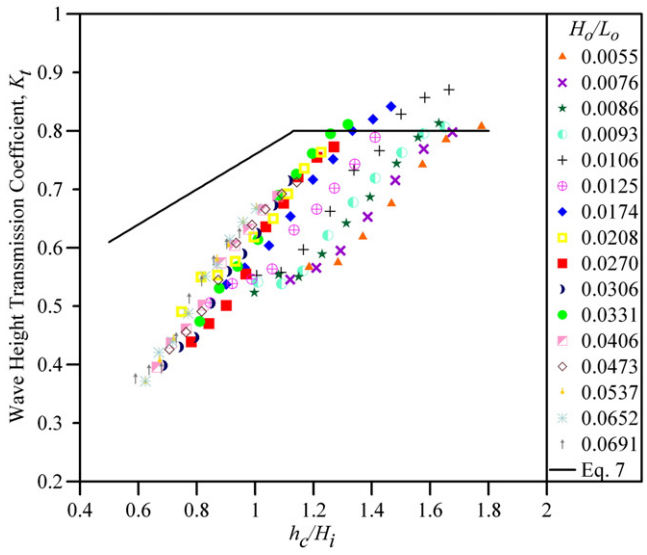


Fig. 10. Measured wave height transmission coefficients as a function of the relative water depth over the reef crest. The solid line represents the values computed by Van der Meer and Angremond (1992) for submerged breakwaters.

Ultimately there will be no flow over the crest and conditions are those of an emergent planar beach, characterised by swash motions as the bores formed by breaking waves propagate shorewards and collapse on the beachface.

3.3. Wave transmission

Knowledge of the wave transmission characteristics of submerged reefs and bars is essential for calculation of nearshore currents, beach evolution and impact forces on shoreline structures.

Measurements were carried out for this purpose with 16 regular wave cases with deepwater wave steepnesses in the range $0.0055 < H_o/L_o < 0.0691$. Each of these was run for 9 different water levels, with the depth over the reef crest h_c in the range 62 mm to 102 mm. Each test was repeated three times.

Wave transmission past both submerged and low emergent breakwaters has been examined by many researchers including Ahrens (1987), Seelig (1980), Van der Meer (1990) and Allsop (1983). The transmission coefficient K_t (the transmitted wave height divided by the incident wave height) was found to be influenced by the incident wave conditions, the submergence of the breakwater crest h_c, the relative crest width of the structure B/L and the nominal diameter of the armourstone used for breakwater construction. Van der Meer and Angremond (1992) later reanalysed a large amount of existing data from irregular wave tests by Ahrens (1987), Daemrich and Kahle (1985), Powell and Allsop (1985), Seelig (1980), Van der Meer (1990) and Allsop (1983), neglecting the influence of the armour size and crest width. Though the data was quite scattered, the following formulae for wave transmission over submerged and low emergent breakwaters were proposed:

$$\begin{aligned} -2.0 < \frac{-h_c}{H_i} < -1.13, \quad & K_t = 0.80; \\ -1.13 < \frac{-h_c}{H_i} < 1.2 \quad & K_t = 0.46 + 0.3\frac{h_c}{H_i}; \\ 1.2 < \frac{-h_c}{H_i} < 2.0, \quad & K_t = 0.10. \end{aligned} \tag{7}$$

An updated analysis of existing data was completed more recently by Friebel and Harris (2004), however this analysis showed a dependence on the relative crest width of the breakwater and so is not easily comparable to the present results where the crest width $B=0$. The formulae suggested by Van der Meer and Angremond (1992) (Eq. (7)) are plotted in Fig. 10 along with the present results, which lie in the range $0.37 < K_t < 0.87$. The results for waves with high deepwater steepnesses ($0.0174 \le H_o/L_o \le 0.0691$) collapse approximately on to a single line, suggesting that in these conditions the energy transmission depends only on the ratio of the water depth at the reef crest to the offshore wave height. However this trend does not hold for less steep waves, and when $H_o/L_o < 0.0174$ the transmission coefficient reduces with decreasing steepness for a given relative water depth h_c/H_i.

Present values of K_t are predominantly lower than those for submerged breakwaters, particularly for cases of low reef crest submergence. A reasonable explanation is that the more gentle gradient of the submerged reef allows the waves to shoal and break by depth limited conditions. Waves breaking on a submerged rubble mound structure with a seaward gradient typically of the order 1:2 have no opportunity to shoal prior to breaking and tend to surge over the crest with a modified breaker type such as a two-step breaker or drop-type breaker (Takikawa et al., 1997) leading to higher transmission.

3.4. Wave reflection

Reflection coefficients K_r were calculated for each of the wave cases presented in the previous section using the method described by Isaacson (1991) where the reflection coefficient is the ratio of reflected and incident wave heights. An initial baseline test demonstrated that the absorbing foam beach installed at the end of the flume had a reflection coefficient in the range 0.8% to 8.5% with the higher values of K_r corresponding to waves with relatively low deepwater steepness ($0.0055 < H_o/L_o < 0.0125$).

The values of K_r measured with the reef in place are plotted as a function of the relative water depth over the reef crest in Fig. 11. It is seen that there is a general trend of increasing reflection coefficient with reducing deep water wave steepness, in agreement with processes known for waves on emergent slopes as well as with the data of Smith and Kraus (1990) for a submerged bar. In addition there is a strong dependence of the reflection coefficient on the relative reef submergence, with the value of K_r becoming larger as the water depth is increased. This surprising result conflicts with the measurements of Ahrens (1987) on reef breakwaters. A possible explanation, consistent with an argument of Battjes (1974), is that over the range of values

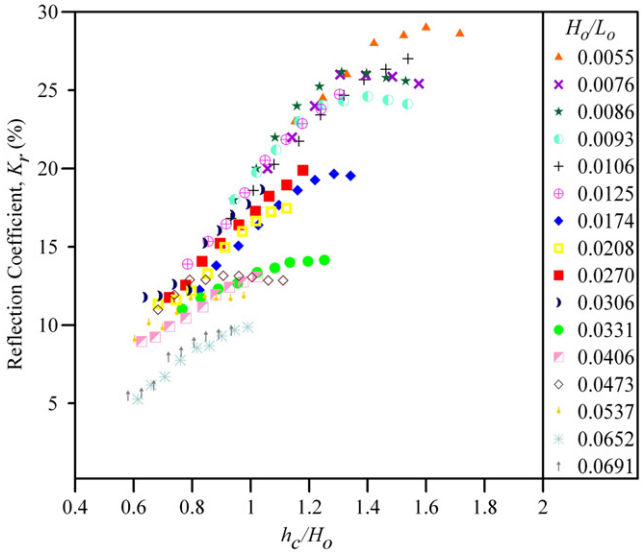


Fig. 11. Measured reflection coefficients as a function of relative water depth over the reef crest.

under examination, as the water depth decreases, more energy is removed from the energy budget through breaking as shown in Fig. 10. Consequently less wave energy is available to be reflected from the reef structure. It is noted that the trend of increasing reflection coefficients with increasing relative crest depths appears to reach a maximum and begin to decrease at higher values of h_c/H_o. This is where the water depth approaches a level at which the waves no longer break.

4. Conclusions

The majority of existing knowledge of shallow water wave breaking has been obtained from laboratory experiments conducted on emergent, plane beaches. In reality, breaking frequently occurs on submerged offshore bars or reefs and it is important to understand the differences between these two situations. In particular, it was the aim of the current study to examine the effect of varying relative crest submergence on the properties of waves breaking over a submerged reef and to this end, a series of laboratory experiments was completed in a wave flume. The results demonstrated that the relative submergence of the reef crest has a significant effect on the nature of breaking and the main findings are listed below.

1. The values of the breaker depth and breaker height indices are comparable to those found on emergent, plane slopes and do not appear to be strongly dependent on the relative submergence of the reef crest.
2. In contrast to previous results on emergent slopes, the breaker depth index was found to be nearly independent of wave steepness, with a value of approximately 0.85 for the present experiments.
3. Breaker type is strongly influenced by the water level over the reef crest. Consequently the classification of breaker type based on the surf similarity parameter suggested by Battjes (1974) for emergent slopes is unsuitable for the description of waves breaking on submerged structures in its current form.
4. An increase in the amount of energy dissipated during the breaking event was observed as the wave breaking intensity increases and the breaker generates increased levels of turbulence, air entrainment and splash.
5. An increase in the velocity of the return flow over the crest was measured as the relative submergence of the reef is reduced and this is thought to be responsible for the observed variation in breaking intensity. It was noted that as the relative crest submergence is reduced, the conditions approach those of an emergent beach. However it is suggested that the nature of the flow over a submerged reef and emergent beach is fundamentally different and a detailed analysis of these differences would constitute an interesting future line of research.
6. The amount of reflection from the reef face becomes greater as the water depth over the reef is increased for waves breaking offshore of the reef crest. It is noted that the trend of increasing reflection coefficients with increasing relative crest depths appears to reach a maximum and begin to decrease at high values of h_c/H_o where the water depth approaches a level at which the waves no longer break.
7. The wave height transmission coefficient decreases as the relative submergence of the reef crest is reduced. The measured values of K_t are predominantly lower than those previously measured for submerged breakwaters, suggesting that a submerged reef with a relatively low seaward gradient is more efficient at dissipating wave energy through wave breaking than a steep sided breakwater.

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